Final Project
Computational Statistics

GROUP 16

Giacomo Puglielli, Jacopo Favaro, Davide Pastori

Bocconi
Università Commerciale
Luigi Bocconi

MSc Data Science & Business Analytics
a.y. 2025-2026

1 Introduction

This project addresses two regression problems using the framework of Generalized Linear Models (GLMs), comparing parameter estimates obtained via Maximum Likelihood Estimation (MLE) and Bayesian inference. Since GLMs rarely yield closed-form solutions, both approaches rely on iterative procedures. MLE estimate parameters by maximizing the log-likelihood using Fisher scoring method while Bayes estimation is based on the posterior distribution, which combines the likelihood with prior beliefs. Because the posterior is not analytically solvable in general, we approximate it using Markov Chain Monte Carlo (MCMC) methods, which sample from the unnormalized posterior. Both approaches provide estimates and measures of uncertainty, offering complementary perspectives when exact solutions are unavailable.

2 Frequentist Approach

2.1 Equivalence between Newton–Raphson and Fisher Scoring in GLMs under canonical link

Let $Y_1, \dots, Y_n$ be independent random variables belonging to the exponential family, with density (or pmf)

$$f(y_i \mid \theta_i, \phi) = \exp\left\{ \frac{y_i\theta_i - b(\theta_i)}{a(\phi)} + c(y_i, \phi) \right\}.$$

The log-likelihood is

$$\ell(\beta) = \sum_{i=1}^{n} \frac{y_i\theta_i - b(\theta_i)}{a(\phi)} + \sum_{i=1}^{n} c(y_i, \phi).$$

In a Generalized Linear Model, the linear predictor is $\eta_i = x_i^\top \beta$, where $x_i \in \mathbb{R}^{p\times 1}$ is the vector of covariates and $\beta \in \mathbb{R}^{p\times 1}$ is the parameter vector.
Under the *canonical link* we have

$$\theta_i = \eta_i = x_i^\top \beta.$$

Score function. Differentiating the log-likelihood with respect to β, since y_i does not depend on β "directly", by the chain rule

$$\frac{\partial}{\partial \beta}(y_i\theta_i) = \frac{\partial y_i}{\partial \theta_i}\frac{\partial \theta_i}{\partial \eta_i}\frac{\partial \eta_i}{\partial \beta}. = y_i \frac{\partial \theta_i}{\partial \eta_i}\frac{\partial \eta_i}{\partial \beta}.$$

Since we have $\theta_i = \eta_i$, then $\partial\theta_i/\partial\eta_i = 1$ so $\partial(y_i\theta_i)/\partial\beta = y_i\, \partial\eta_i/\partial\beta$. The same applies to $\partial b(\theta_i)/\partial\beta$
Thus

$$\frac{\partial \ell}{\partial \beta} = \sum_{i=1}^{n} \frac{1}{a(\phi)}\big(y_i - b'(\theta_i)\big)\frac{\partial \eta_i}{\partial \beta}.$$

Since $\eta_i = \theta_i$ and $\theta_i = x_i^\top\beta$, it follows that $\partial\theta_i/\partial\beta = x_i$. Moreover, we recall that for exponential family distributions, $\mu_i = \mathbb{E}(Y_i) = b'(\theta_i)$ and $b''(\theta) = (a(\phi))^{-1}Var(Y_i)$. Therefore, the score function can be written as

$$U(\beta) = \frac{1}{a(\phi)}\sum_{i=1}^{n} x_i(y_i - \mu_i) = \frac{1}{a(\phi)}X^\top(y - \mu) \quad \text{where} \quad X \in \mathbb{R}^{n\times p}, y, \mu \in \mathbb{R}^{n\times 1}$$

Hessian matrix. If the link is canonical, $\theta_i = \eta_i = x_i^\top\beta$ and therefore

$$\frac{\partial^2 \theta_i}{\partial\beta\partial\beta^\top} = 0,$$

so the Hessian does not contain terms depending on $(y_i - \mu_i)$. For non-canonical links, θ_i is nonlinear in β and an additional term proportional to $(y_i - \mu_i)$ appears in the observed Hessian.
Indeed, in this case $\eta_i = \theta_i$; therefore the observed Hessian is a $p \times p$ matrix like this

$$\nabla^2_\beta \ell(\beta) = \frac{1}{a(\phi)}\sum_{i=1}^{n} x_i \left(-\frac{\partial \mu_i}{\partial \beta}\right)^\top.$$

Since $\mu_i = b'(\theta_i)$ and $\theta_i = x_i^\top \beta$,

$$\frac{\partial \mu_i}{\partial \beta} = b''(\theta_i)\frac{\partial \theta_i}{\partial \beta} = b''(\theta_i) x_i.$$

Hence,

$$\nabla^2_\beta \ell(\beta) = -\frac{1}{a(\phi)} \sum_{i=1}^{n} b''(\theta_i)\, x_i x_i^\top = -\frac{1}{a(\phi)} X^\top W X, \quad \text{where} \quad W = \mathrm{diag}\big(b''(\theta_1), \ldots, b''(\theta_n)\big)$$

Equivalence. Therefore, the observed Hessian coincides with its expectation where the expectation is taken with respect to the joint distribution of $(Y_1, \ldots, Y_n)$ under the model with parameter β. Again, this equality holds because the canonical link implies $\partial^2 \theta_i / \partial \beta \partial \beta^\top = 0$.

$$\mathcal{I}(\beta) = -\mathbb{E}\big[\nabla^2_\beta \ell(\beta)\big] = -\nabla^2_\beta \ell(\beta),$$

As a consequence, the Newton–Raphson update coincides with the Fisher scoring algorithm, that is

$$\beta^{(t+1)} = \beta^{(t)} - \big[\nabla^2_\beta \ell(\beta^{(t)})\big]^{-1} U(\beta^{(t)}) = \beta^{(t)} + \mathcal{I}(\beta^{(t)})^{-1} U(\beta^{(t)})$$

Probit regression In this project we only consider GLMs with canonical link regression). A well-known example of GLMs with non-canonical link is the probit regression for binary data, where

$$\mu_i = \Phi(\eta_i), \qquad \eta_i = x_i^\top \beta,$$

with Φ denoting the standard normal distribution function. In this case, the Fisher and Newton–Raphson have different update rules, anchored to the difference between the expected and observed Hessian.

2.2 Algorithm Implementation

In Section 2.1 we derived, for both logistic and Poisson regression under a canonical link, the Newton (equivalently Fisher scoring) update in the common form

$$\beta^{(t+1)} = \beta^{(t)} + (X^\top W^{(t)} X)^{-1} X^\top \big(y - \mu^{(t)}\big), \qquad W^{(t)} = \mathrm{diag}(w_i^{(t)}),$$

where $\eta^{(t)} = X\beta^{(t)}$ and the weights satisfy $w_i^{(t)} = Var(Y_i)$ because both in Bernulli and Poisson $a(\phi) = 1$. We now rewrite this update as a linear system for implementation purposes. Multiplying both sides by $X^\top W^{(t)} X$, using $\eta^{(t)} = X\beta^{(t)}$, and defining $z^{(t)}$:

$$z^{(t)} = \eta^{(t)} + \big(W^{(t)}\big)^{-1} \big(y - \mu^{(t)}\big),$$

we obtain

$$\beta^{(t+1)} = \Big(X^\top W^{(t)} X\Big)^{-1} X^\top W^{(t)} z^{(t)}$$

After solving this system, we obtain at each iteration the updated parameter vector, denoted in the code as β_{new}.

3 Bayesian Approach

We adopt a Bayesian approach to inference, in which uncertainty on the regression coefficients β is quantified through the posterior distribution. The posterior combines the information provided by the data with prior beliefs on the parameters and is given, up to a normalizing constant, by

$$p(\beta \mid Y) \propto p(Y \mid \beta)\, p(\beta)$$

Prior distributions We assign independent Gaussian priors to all regression coefficients, centered at zero. [1] The different prior scales reflect the different effects induced by the logit and log links, with larger variability allowed in the logistic model and more conservative priors in the Poisson case. and with fixed variance equal to 4 and 1 respectively in the logistic and in the Poisson regression. Larger variances of 9 and 4 are used for the intercepts to allow additional flexibility in capturing baseline effects not explained by the covariates.

[1] `reports` was the most plausible candidate for a negative prior mean: more credit reports are often interpreted as a negative signal. At the same time, the effect is ambiguous because some lenders may view such customers as more profitable (e.g., by charging higher interest rates). Given this uncertainty, we kept zero-centered priors for `reports` and for all other coefficients (which had even lower reasons to be changed) and let the data drive inference.

Likelihood The log-likelihoods were computed with respect to the formula of log-likelihood for exponantial family distribuitions, as presented in Section 2.1. For the Credit Card dataset we used the Bernoulli, while for the Hospital dataset the Poisson.

Algorithm implementation We use a Metropolis–Hastings within Gibbs algorithm to sample from the posterior distribution. Starting from an initial parameter vector, the algorithm updates the coefficients one at a time at each iteration, instead of proposing a joint update. For each coefficient, a new value is proposed via a random-walk step drawn from a univariate normal distribution centered at the current value, while all other coefficients remain fixed. The proposed move is then accepted or rejected according to the Metropolis–Hastings acceptance rule:

$$\alpha = \min\left\{1, \frac{\pi(\beta^{(t)} \mid y)}{\pi(\beta^{(t-1)} \mid y)}\right\}$$

After all coefficients have been updated once, the resulting vector is stored as the next state of the chain. Updating one coefficient at a time prevents a single bad proposal from blocking the whole vector, leading to lower autocorrelation and faster exploration. Multivariate MH is still valid, but the component-wise approach proved more efficient here (we tested both).

Initial value of β. The chain is initialized at β_{GLM}, which lies in a high-probability region of the posterior since, by the Bernstein–von Mises theorem, the posterior is asymptotically centered at the MLE. This choice significantly reduces the burn-in period; however, alternative starting points were also tested and did not affect convergence.

Proposal variance. The proposal step controls the trade-off between acceptance and exploration. For this reason, we tune the proposal variance by minimizing the autocorrelation of the worst-performing component, trying to ensure efficient exploration of all coefficients.[2]

Acceptance probability. The Metropolis–Hastings acceptance probability is constructed to ensure that the posterior distribution is an invariant distribution of the Markov chain. Under suitable regularity conditions, this invariant distribution is also the unique stationary distribution.

1) Existence of an invariant distribution. The Markov chain admits an invariant distribution $\pi(\beta)$ when it satisfies the detailed balance condition for symmetric proposal distributions:

$$\pi(\beta_a)\, T(\beta_a \to \beta_b) = \pi(\beta_b)\, T(\beta_b \to \beta_a) \implies \pi(\beta_a)\, \alpha(\beta_a, \beta_b) = \pi(\beta_b)\, \alpha(\beta_b, \beta_a), \qquad \beta_a \neq \beta_b,$$

where $T(\beta_a \to \beta_b)$ denotes the probability of moving from β_a to β_b in one iteration.

2) Identification of the invariant distribution with the posterior. Substituting the Metropolis–Hastings acceptance rule into the detailed balance condition yields[3]

$$\pi(\beta_a)\, \min\left\{1, \frac{p(\beta_b \mid y)}{p(\beta_a \mid y)}\right\} = \pi(\beta_b)\, \min\left\{1, \frac{p(\beta_a \mid y)}{p(\beta_b \mid y)}\right\}.$$

To verify this equality, consider the two possible cases:
A) $p(\beta_b \mid y) \geq p(\beta_a \mid y)$
B) $p(\beta_b \mid y) < p(\beta_a \mid y)$

$$\text{(A)} \quad \pi(\beta_a) \cdot 1 = \pi(\beta_b) \cdot \frac{p(\beta_a \mid y)}{p(\beta_b \mid y)} \qquad\qquad \text{(B)} \quad \pi(\beta_a) \cdot \frac{p(\beta_b \mid y)}{p(\beta_a \mid y)} = \pi(\beta_b) \cdot 1$$

Both equalities hold if and only if $\pi(\beta) = p(\beta \mid y)$. Hence, under the MH acceptance rule, the posterior distribution is invariant for the Markov chain. Moreover, since the chain is irreducible[4] and aperiodic[5], the invariant distribution is unique. Therefore, the posterior coincides with the unique stationary distribution of the Markov chain.

[2] We find a locally optimal step size based on our initial β, which may not be a global best step. See the code...
[3] $p(\beta \mid y)$ denotes the posterior density of the parameter vector β given the observed data y, defined up to a constant.
[4] Irreducibility follows from the Gaussian random–walk proposal, which has full support on $\mathbb{R}^p$, allowing any state to be reached with positive probability.
[5] Aperiodicity follows from the possibility of rejection, which induces self–transitions.

4 Results

4.1 Estimates

Table 1: Logistic Regression: Comparison MLE and Bayes Estimator

	MLE			Bayesian			
Covariates	**CI 2.5%**	**FisherScoring**	**CI 97.5%**	**2.5% cred.**	**Post mean**	**97.5% cred.**	**Acc. Rate**
reports	-2.7274	-2.3556	-1.9837	-2.7412	-2.3643	-2.0053	0.5703
intercept	1.0579	1.2998	1.5417	1.0699	1.3095	1.5581	0.3679
active	0.6013	0.8339	1.0665	0.5986	0.8294	1.0624	0.3947
income	0.1700	0.3832	0.5964	0.1828	0.3925	0.6144	0.4144
dependents	-0.4714	-0.3022	-0.1331	-0.4706	-0.3031	-0.1314	0.3599
majorcards	0.0508	0.1953	0.3398	0.0458	0.1950	0.3451	0.3362
selfemp	-1.3239	-0.7573	-0.1908	-1.2947	-0.7386	-0.1521	0.7268
owner	0.0859	0.4783	0.8706	0.0954	0.4792	0.8704	0.5353
age	-0.3176	-0.1269	0.0638	-0.3119	-0.1241	0.0671	0.3685
months	-0.1472	0.0338	0.2148	-0.1362	0.0367	0.2273	0.3668

Table 2: Comparison of MLE and Bayesian estimates

	MLE			Bayesian			
Covariates	**CI 2.5%**	**FisherScoring**	**CI 97.5%**	**CI 2.5%**	**Post mean**	**CI 97.5%**	**Acc. Rate**
intercept	1.8119	1.8310	1.8501	1.8119	1.8310	1.8499	0.4441
procedure	0.5457	0.5643	0.5829	0.5450	0.5639	0.5826	0.4349
type	0.0779	0.0943	0.1108	0.0771	0.0943	0.1107	0.4355
gender	-0.0654	-0.0489	-0.0325	-0.0654	-0.0488	-0.0324	0.4401
age	0.0267	0.0432	0.0597	0.0266	0.0434	0.0601	0.4555
died	-0.0571	-0.0412	-0.0254	-0.0578	-0.0416	-0.0261	0.4257

4.2 Interpretation

Logistic Regression In logistic regression, each coefficient β_j represents the effect of a unitary increase in the corresponding covariate x_j, keeping everything else fixed, on the log-odds of the binary outcome. Defining Z_i^j as the observation vector X_i with covariate j increased by one unit, we have:

$$\beta_i = log(\frac{\Omega(z_i^j)}{\Omega(x_i)}) \quad \text{or} \quad e^{\beta_i} = \frac{\Omega(z_i^j)}{\Omega(x_i)} \quad \text{where} \quad \Omega(X_i) = \frac{\mathbb{P}(Y_i = 1 \mid X_i)}{1 - \mathbb{P}(Y_i = 1 \mid X_i)}$$

Positive coefficients indicate an increase in the probability of the event, while negative coefficients suggest a decrease.
Analyzing the coefficients in Table 1, the most statistically significant covariates are the number of derogatory reports, the intercept, and the number of active credit accounts. These variables capture the main drivers of credit-card approval. As expected, derogatory reports have a strong negative impact: each additional report sharply lowers the odds of approval, with an odds ratio of about 0.1, indicating that lenders tend to view applicants with multiple negative records as considerably less trustworthy. The intercept's significance points to a relatively high baseline probability of approval at the reference levels of the covariates. Active credit accounts, instead, have a positive effect, suggesting that applicants who already manage credit responsibly are viewed as more reliable borrowers.

Poisson Regression In Poisson regression, each coefficient β_j represents the effect of the corresponding covariate x_j on the log of the expected count. Specifically, defining Z_i^j as in 4.2 we have:

$$\beta_j = \log\left(\frac{\lambda(Z_i^j)}{\lambda(X_i)}\right) \quad \text{or} \quad e^{\beta_j} = \frac{\lambda(Z_i^j)}{\lambda(X_i)}$$

Positive coefficients indicate an increase in the expected number of events, while negative coefficients suggest a decrease.

Analyzing the estimated coefficients reported in Table 2, we gain insight on how the covariates influence the expected number of events. In a Poisson regression framework, each coefficient represents the effect of the corresponding covariate on the log of the expected count. The intercept reflects the baseline expected count for a reference patient, and its large positive value indicates a relatively high baseline rate. Among the covariates, the procedure type has a strong positive effect: suggesting that CABG procedure, as opposed to PTCA, substantially increases the expected count of the hospital days. Emergency or urgent admissions also exhibit a positive association with the outcome, suggesting higher expected counts compared to elective cases.

4.3 MCMC Diagnostic

To assess the quality of the MCMC results, we verified three key aspects: convergence of the algorithm, convergence to the posterior distribution, and computational efficiency. In both models, convergence was evaluated through trace plots, which show that after burn-in the sampled values fluctuate around a constant horizontal level centered at the posterior mean, indicating that the chain has reached stationarity and is sampling from the target posterior distribution. Finally, efficiency was assessed through autocorrelation functions (ACF): the observed decay of autocorrelation to zero within 15–20 lags indicates weak dependence between successive samples and effective exploration of the parameter space. This conclusion is further supported by acceptance rates in a reasonable range (Logistic = 44.41% and Poisson = 37.13%), confirming that the proposal distribution achieves a good balance between exploration and rejection.

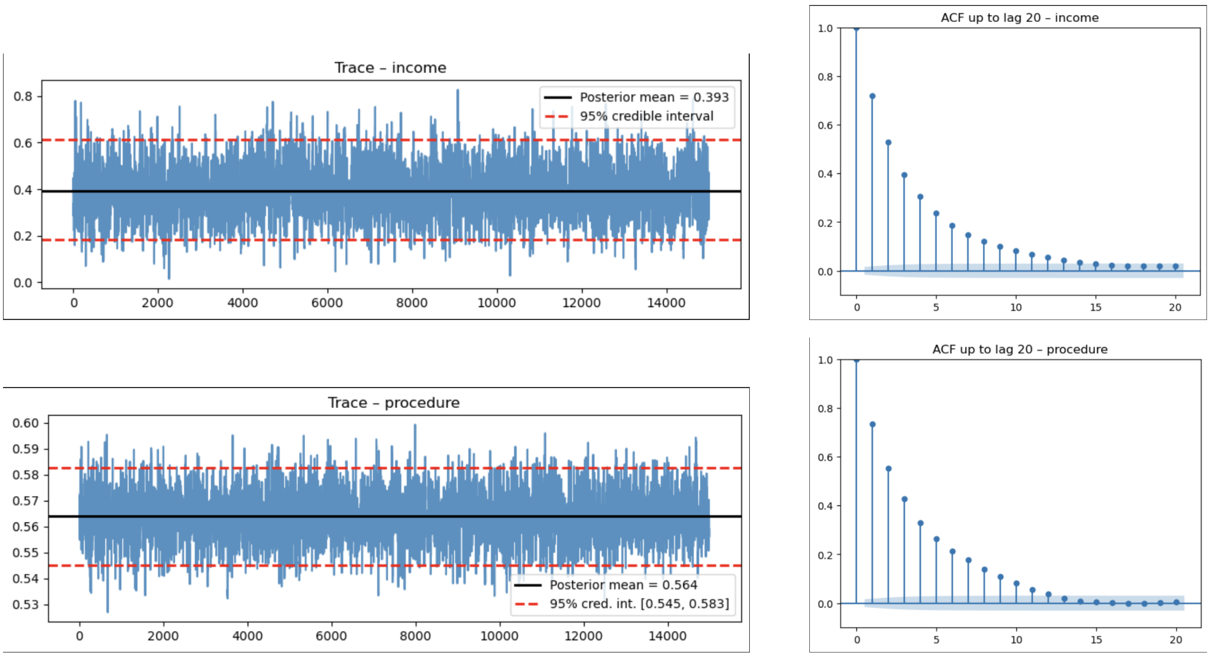

Figure 1: Traceplots and autocorrelation functions for selected coefficients in bayesian regressions.

5 Conclusion

In both regressions, frequentist and Bayesian estimates are very similar due to the use of weakly informative priors, which allow the data to dominate the posterior. With strongly informative priors, Bayesian estimates would differ from the MLE, illustrating the broader scope of the Bayesian approach which allows prior beliefs to influence inference when appropriate (although the Bernstein–von Mises theorem ensures asymptotic agreement as the sample size grows to infinity). From a computational perspective, the component-wise Metropolis–Hastings proved more efficient than the multivariate version. This difference is likely due to high dimensionality (especially in the logistic model), where joint proposals increase the probability that at least one poorly proposed coefficient leads to rejection of the entire vector, and therefore slow the chain.